SQL Commands

- Data Definition Language (DDL)
  - Create
  - Alter
  - Drop
  - Truncate
  - Rename
- Data Manipulation language (DML)
  - Select
  - Insert
  - Update
  - Delete
- Data Control Language (DCL)
  - Grant
  - Revoke
- Transaction control language
  - Commit
  - Rollback
  - Savepoint
- Contraints
  - Primary Key
  - Foreign key
  - Check
  - Unique
  - Default
  - Not Null

asit properties

These are used in PDL and DML

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Normalization:

It has three operations.
1- Insertion Anomaly
2- Deletion Anomaly
3- Updation Anomaly.

~~All Normal Forms~~:
we use with (Group by.)
→ But when we are writing nested query then we can use (Where) in out query.
→ Use of (Exist and Not Exist)
We use Exist and Not Exist in Corelated nested query. It returns Ans in T/F con
→ We take one value from outer query

→ We use equal (=) to compare one value but if we want to compare more than one value then we use IN.
like 2 IN (1,3,4,5)
→ (Where) use for all table means compare with whole table.
→ (Group by) and (WHERE) will not work together.
→ we use (HAVING) with (Group by)
→ We can use agregate function with select in case of Group by operation, but we can't use an other command.
→ We always use the same command with (Select) that we

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DDL (Data Definition Language)

* Create table
* Alter table
* Drop table
* Truncate table
* Rename

Step-1 (Structure)

```
(Example)
(
  Colum1 name datatype,
  Colum2 name datatype;
  Colum3 name datatype      (no comma)
);
desc table_name;
```

Result will be of create tabe

column	Null?	Type
ID	-	NUMBER
NAME	-	VARCHAR2(10)
SALARY	-	NUMBER(100)

Step-2 Right Exam

1- Create table :-

```
create table emp
(
id int,
name varchar2(10),
salary number(10),
);
desc emp;
```

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SQL Query

Delete

(DML) command

Q→ Delete frome table name
Delete from Student;

Student

ID	Name
1	A
2	B

- It detele ~~ed~~ rows from table. one by one.
- After using query the rows will delete one by one but table will not delete.

Q- Delete from Student
Where ID = 1;

→ It is Slower. It has log.

Drop

(DDL) ~~command~~

Q- Drop table Student;

- It will delet whole table and it is very dangerous fom of DDL.
- After using query table will be disappear.

→ We can't rollback it.

Truncate

(DDL).

Q- Trancate Student;

- Desc Student;

ID	name

- It also detete rows from table but at one time.
- It has no option.
- It is faster
- If we want to Rollback ~~Student~~ (data) it will not possible.

→ It has no log

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This is Second Command

Alter (DDL)

Q→ Alter table Student add address varchar(30);

Uses of Alter Command

1- Add column/s
2- Remove Coulumn/s
3- Modify datatype
4- Modify datatype length.
5- Add constraints
6- Remove constraints
7- Rename colum/table.

Full commands

```
Create table employee
(
 id int,
 name varchar(10)
);
alter table employee add address varchar(10);
desc employee
alter table employee drop column address;
alter table employee modify id varchar(10);
alter table employee rename column id to
roll_no;
alter table employee rename to emp11;
```

then desc emp11.

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alter table emp11 add primary key (roll_no)

Difference

Rename :- UPDATE (DML)	Alter (DDL)
→ We use Update command for change in data.	→ If we want change in table structure we use Alter command.
Exam: Update Emp set Salary = Salary * 2;	→ Exam: if we wan add data in already existing table we use Alter command.
→ We can use conditions in it.	→ For delete column
→ We can change name, salary, mail etc.	→ Datatype change
	Back Pg

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Constraints in SQL

Constraints means using condition and a user can add data according to these conditions. And we used conditions on attributes or columns.

Types of Constraints:

1- Unique
2- Not null
3- Primary = Unique + Not null.
4- Check constraints
5- Foreign key
6- Default

Check Constraints:

Check means we have to check value before put it in column.

Example:

check (age > 18)

if user put 10 value it will give error but in case of 20 it should be correct.

Unique:

In this contraints duplicate data is not allowed. in a column to which we selected as unique.

Not null:

If we select a column as a not null a user will not fill it and can leave this colum on the other hand if we take this not null a user must have to fill this column. Otherwi it will show error.

Null means empty colum - not 0

Primary Key:

It should be unique and not null. it is imp contraints.

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Foreign Key:

It is also used for the refrential integrity. It is also very important part of contraints. Like if we have two tables if these tables take refernce from each other. then this reference should be proper.

Default:

In this constraint we use any value as a defualt and after defalt it column data will select this default value by itself. Like:

Salary int default 10000

→ In this case if we don't put value in column default command will automatically call default value.

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SQL Queries and Subqueries

Emp

E-ID	E-name	Dept	Salary
1	Ram	HR	10000
2	Amrit	MRKT	20000
3	Ravi	HR	30000
4	Nitin	MRKT	40000
5	Varun	IT	50000

E-name

(1) ⇒ Write a SQL Query to display maximum Salary from Emp. Table.

(2) ⇒ Write a SQL Query to display Employee name who is taking maximum Salary

(*) For print max/min salary we use agree-gate functions.

Query of (1)

```
Select Max(Salary) from Emp;
```

Output

⇒ 50000

Query of (2) in this we use sub-Q

```
Select E-name from Emp where Salary = (Select Max(Salary) from emp);
```

outer Query

inner Query

Output

Varun

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Going Same Table Queries.

(3) Write a SQL to display 2nd highest Salary from Emp Table?

Query

```
Select Max(Salary) from
    Emp where Salary<>(Select Max(Salary) from Emp);
```

Answer:

```
Select Max(Salary) from
    Emp where Salary<> (Select
        Max (Salary) from Emp;
```

Output

40000

(4) Write a SQL Query to display Employee name who is taking Second highest Salary?

Query

```
Select E-name from Emp
    Where Salary =
(Select Max(Salary) from
    Emp where Salary <> (Select
        Max (Salary) from Emp);
```

Output

Nitin

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In this query we use ↓ by Group function

Q5: Write a query to display all the dept names along with no. of Emp working in that?

Query

~~Select dept from Emp group by dept;~~ ✗

✓ Select dept, count(*) from emp group by dept;

(option) or dept

Q6: Write a query to display all the dept names where No. of emps are less than 2.

Query

```
Select dept from Emp group by dept
                    having Count(*) < 2;
```

Output

IT

Q6

In case of finding name of Emp ~~whose~~ where no. of emps are less than 2.

~~Ist~~ Query

```
Select E-name from emp
where dept In
(Select dept from Emp group by
dept having count(*) < 2);
```

Output

Varun

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Q:7

Write a query to display highest salary department wise and name of emp who is taking that salary.

Query

```
Select Max(Salary) from Emp group by dept;
```

Output

30000
40000
50000

Pro Query according to emp-name

```
Select E-name from Emp where Salary
In (Select Max(Salary) from Emp group by dept);
```

Now Output

Ravi, Nitin, Varun

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* means show all attributes
For compairing two values we use equal (=) sign
For compairing more than two values we use In In

Ques : 8

Use of In / Not In

Find Detail of Emp whose address is either Delhi or Chd or Pune?

Query

```
Select * from Emp where
    Address In ('Delhi', 'Chd', 'Pune');
```

Output

1	Ravi	Chd
2	Varun	Delhi
3	Nitin	Pune
5	Ammy	Chd

But in case of Not In result should be

4	Robin	Banglore

Emp

Eid (PK)	E-name	Address
1	Ravi	Chd
2	Varun	Delhi
3	Nitin	~~Banglore~~ Pune
4	Robin	Banglore
5	Ammy	Chd

Project

Eid (FK)	Pid (PK)	Pname	Location
1	P_1	IOT	Banglore
2	P_2	BIG Data	Delhi
3	P_3	Retail	Mumbai
4	P_4	Android	Hydrabad

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Q#9: Use of Exists/ Not exists

Find the detail of Emp who is working on at least one project?

Corelated Nested Query

```
Select * from Emp where
        Eid exists (Select Eid
                from project where
                Emp.Eid = project.Eid)
```

Now Use of Not exists: (name of table)

Query

```
Select * from Emp where
    Eid Not exists (Select Eid from
                project where
                Emp.Eid = project.Eid)
```

Output

* 2, Varun, Delhi

If gives output in the form of true or false.
In case of exist if data is exist in both tables it will return true and vise virsa

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Q#10

Aggregate functions

There are five aggregate functions:

Max, Min, Count, Avg, Sum

Query (For Max value)

```
Select Max(Salary) from Emp;
```

Output

50000

Query (For Min value)

```
Select Min(Salary) from Emp;
```

Output

10000

Query: (For Count)

```
Select Count(*) from Emp;
```

⇒ It find the total number of rows:

Output

6

Emp Table

E-id	E-name	Dept	Salary
1	Ram	HR	10000
2	Amrit	MRKT	20000
3	Ravi	HR	30000
4	Nitin	MRKT	30000
5	Varun	IT	50000
6	Sandy	Testing	Null

~~ne~~ null means no value
It is not 0(zero). means empty

Query:

```
Select count(Salary) from emp;
```

Output: 5

Query:

```
Select Distinct (Count(Salary)) from emp;
```

Output:

4

it will not repeat the va
it will select the unique entities

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Aggregate function

Query (For Sum)

Select Sum(Salary) from Emp;

Output : 140000

Query :

Select Distinct (Sum(Salary)) from Emp;

Output: 110000

Query (For Avg)

Select Avg(Salary) from Emp;

Formula of Avg:

$$Avg(Salary) = \frac{Sum(Salary)}{Count(Salary)} = \frac{140000}{5} = 28,000$$

Output : 28,000

Query :

Select Distinct(Avg(Salary)) from Emp;

Then ; Dis (Avg (Salary)) = $\frac{Distinct(Sum(Salary))}{Distinct(Count(Salary))} = \frac{11000}{4} = 2750$ output

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Correlated Subquery (Synchronized Query)

→ It is a subquery that uses values from outer query.

→ Top to down approach.

Question:

Find all employees detail who work in a department.

Query

```
Select * from Emp where
        exists (Select * from dept where dept.Eid = Emp.Eid);
```

exists is used as connector.

Dept

Did	Dname	Eid (FK)
D_1	HR	1
D_2	IT	2
D_3	MRKT	3
D_4	Testing	4

Emp

Eid	name	Address
1	A	Delhi
2	B	Pune
3	~~A~~	Chd
4	B	Delhi
5	C	Pune ~~Mumbai~~
6	D	Mumbai
7	E	Hyd

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Indexing (Only Sneria)

Index is a list of memory boxes which store data.
It is like we have Index Page in our book and it is the 1st page of our book where we can find required Page No. or retail.
→ It is I/O cost.
→ It actually reduce number of blocks which we count.
→ I/O cost reduced by indexing.
→ In case of DB we send query to System (CPU) and system process it. But data don't store it in CPU. directly store
→ We use our data in Hard disk for permanent storage, and data store in blocks in CPU but these are logical blocks not physical. These blocks are of fixed size.
→ CPU does not intrect directly with Hard disk it will ask to RAM for taking data, and RAM will call data from secondary memory.
→ Data can be sorted and unsorted.

Non Ordered =

In case of unsorted data Hard disk
→ Search data from RAM one by one
If it find data it means Hit.
If don't find data it means Miss.

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